



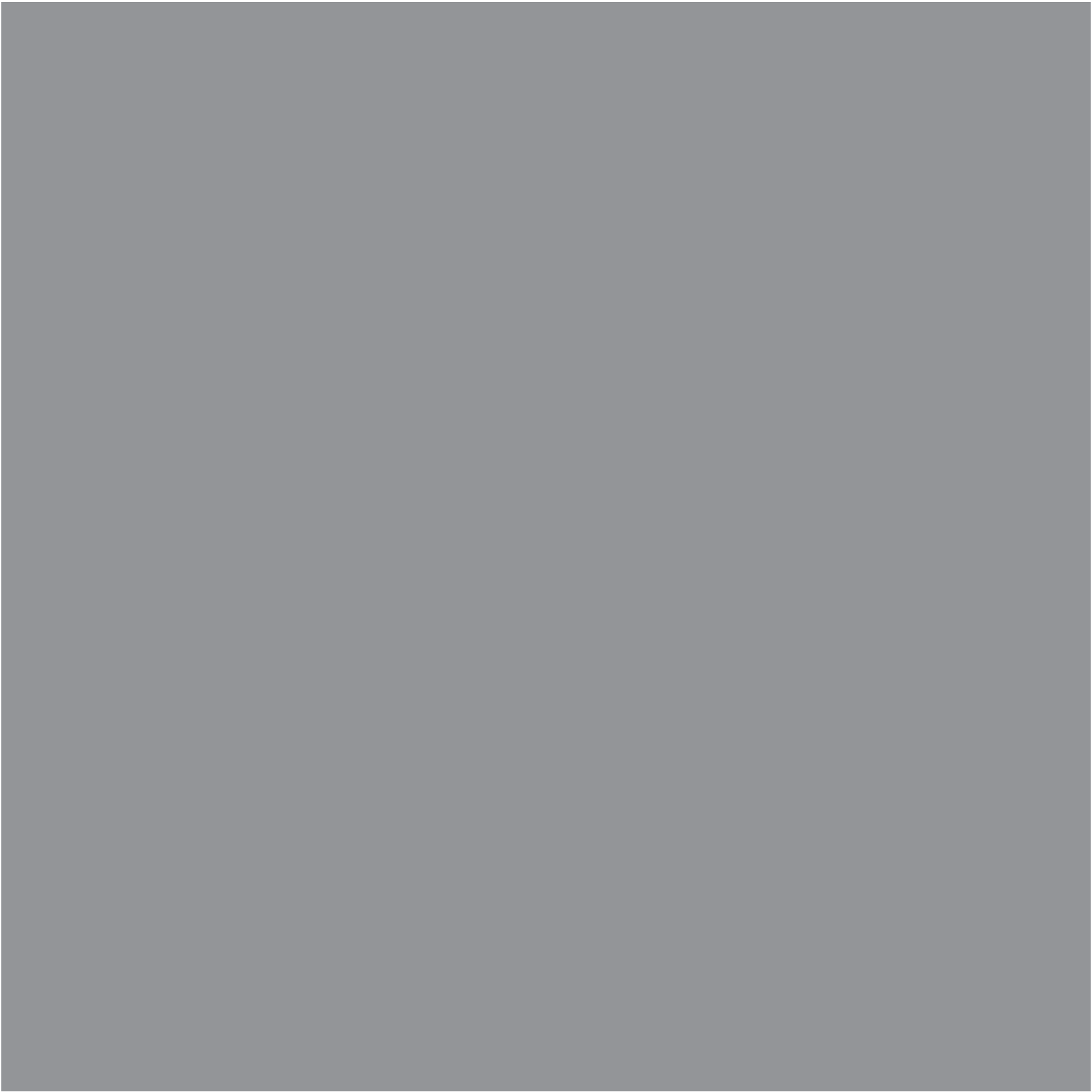
MIJU RAIL

에이치제이스틸 주식회사



에이치제이스틸 주식회사

ILKYUNG INC.



THE WORLD BEST RAIL MANUFACTURER

INTRODUCTION

ELEVATOR GUIDE RAIL

LIGHT RAIL / RAIL

TRACKER

SPECIAL SHAPE



● National Network

Classification		Location		Capacity
Dohwadong Factory	Milling Factory	696/700, Dohwadong, Nam-gu	Rolling Mil	105,000MT
	Machining Factory	693-1, Dohwadong, Nam-gu	L/RAIL Machining	36,700MT
		TEL : 032-864-6000, FAX : 032-864-6006		
Namdong Factory	Machining Factory	645-13, Gojandong, Namdong-gu	E/RAIL Machining	25,700MT
		TEL : 032-815-2691, FAX : 032-811-8524		
Seoul Office		#88-19 Ilkyung Bldg., Daebang-dong, Dong jak-gu, Seoul, Korea		
		TEL : 02-820-2655, FAX : 02-826-6979		

● Export Record

E/RAIL	OTIS-SINGAPORE	The 5th HDB project by Singapore National Housing Corporation
	FUJITEC-JAPAN	High speed rail for the domestic market of Japan (320m/min)
	THYSENKRUPP-BRAZIL	Supply of high speed rail
	OTIS-PANAMA	Skyscraper : Ocean Tower / Vitri Tower in Panama
L/RAIL	SHIMADA-JAPAN	Exclusive contract to SHIMADA SHOKAI
	BIRMINGHAM RAIL-USA	Customized production of feet length rail for USA
	ATLANTIC TRACK-USA	Developed 40lbs rail for USA
	Europe	Developed DIN - S20 for Europe
	Asia	Developed 37kg
	Others	Peru, Chile, Vietnam, Taiwan, Saudi-Arabia, Iran, Indonesia, Colombia, etc..

● Delivery Performance

Classification	Elevator Maker	Builder	Site
E /RAIL	OTIS	Daewoo	Sindorim Techno Mart
		SAMSUNG	Incheon Gansuk KNCH
		POSCO	Songdo 1st World Apartment
		HYUNDAI	Catholic University Hospital
		GS	The City7 Apartment, AIG SIFC
	MITSUBISHI	DAEWOO	Busan Centum Leaders mark
		LOTTE	Hwaseong Dongtan Apartment
		SAMSUNG	Kwacheon Re-construction
		POSCO	KonKuk University Starcity
		GS	Yeouido Gai Apartment
	THYSSENKRUPP	Dusan	Haeundae Dusan We've the Zenith
		DAELIM	Daelim Shinhyun e-Comfort Land
		LOTTE	Apartment in Daegu Jungri
		SAMSUNG	Tower Palace
		POSCO	Haeundae Centum Apartment
		SK	Apartment in Oryukdo
			Vietnam Hanoi Land mark B/D
	HYUNDAI	LOTTE	Hwanghakdong Lotte Castle Apartment
G I-BEAM	Korea Coal Corporation		

● Status of Certification

Type	Standard	Scope of Certification	Certification No.	Certifying Organization
ISO	ISO 9001:2008	Production, sales and additional services of elevator guide rail, light rail, check-ered channel, Gallery I-beam, deck plate, base plate for railroad and channel	QMS-0867	KOREA STANDARD ASSOCIATION
	ISO 14001:2004	Production, sales and additional services of elevator guide rail, light rail, checkered channel, GI-beam, deck plate, base plate for railroad and channel	EMS-0460	KOREA STANDARD ASSOCIATION
KS	KS E 4002	Gallery I-beam, deck plate, base plate for railroad and channel	4380	KOREA STANDARD ASSOCIATION
Q+	Q+ Process	Production, sales and additional services of elevator guide rail	Level 3	OTIS

● Rolling Mill Process



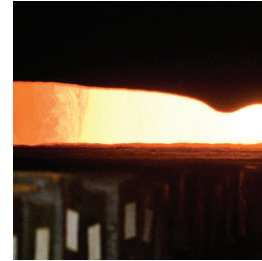
1. Billet



2. Raw Material Inspection



3. Oxygen cutting



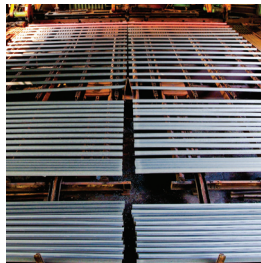
4. Reheating in Furnace



5. Rolling Mill



6. Cutting Off



7. Cooling

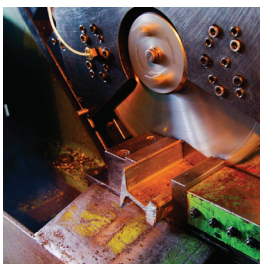


8. Dimension Inspection

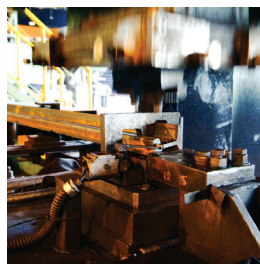


9. Roller Straightening

● Manufacturing Process For Light Rail



1. Cutting Off



2. Punching For Hole



3. Final Inspection
(Mechanical & Chemical Properties)



4. Packing

● Manufacturing Process For E/RAIL



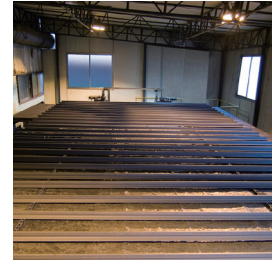
1. Cutting Off



2. Shot Blasting



3. Painting



4. Drying



5. Marking



6. Planer Machining



7. Hydraulic Straightening



8. Straightness Inspection



9. Forming Process



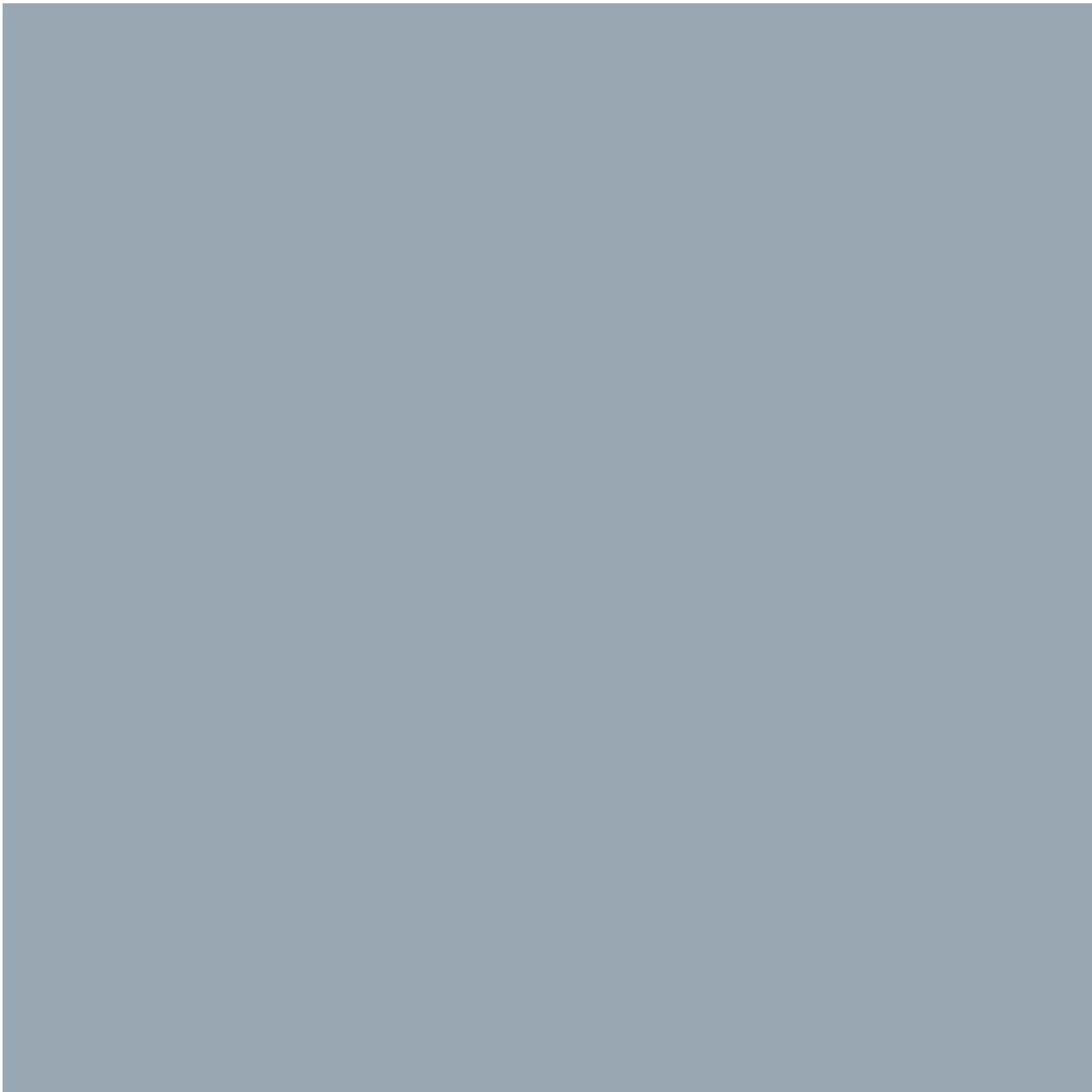
10. Anti-Rust Process



11. Final Inspection
(Dimension & Shape Inspection)



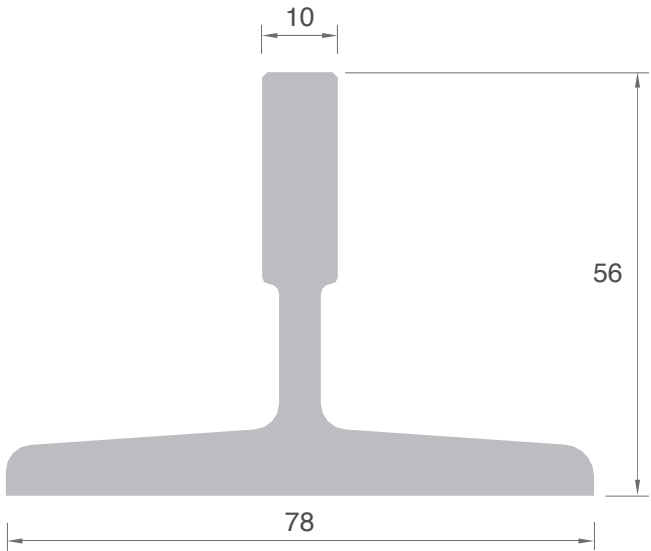
12. Packing



ELEVATOR GUIDE RAIL

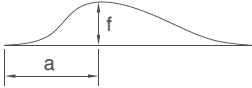
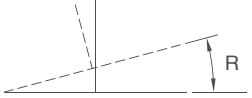


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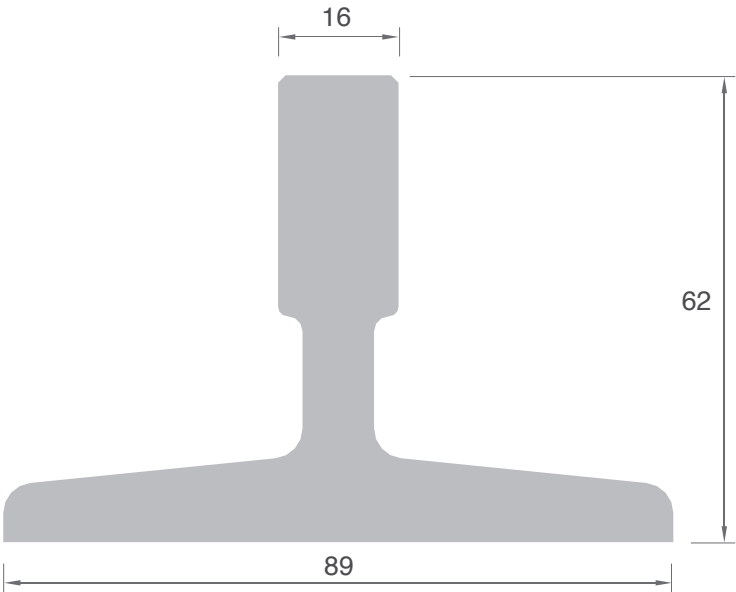


 SPECIFICATION

Unit Weight		7.67 (kg/m)
Standard Length		5,000 (mm)
Roughness	Long	$Ra \leq 3.2 \mu m$
	Trans.	$Ra \leq 6.3 \mu m$

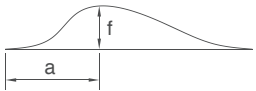
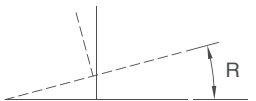
Straightness		f/5m	2.0mm
		f/0.5m	0.2mm
Twist		R	30'/m

Real Size

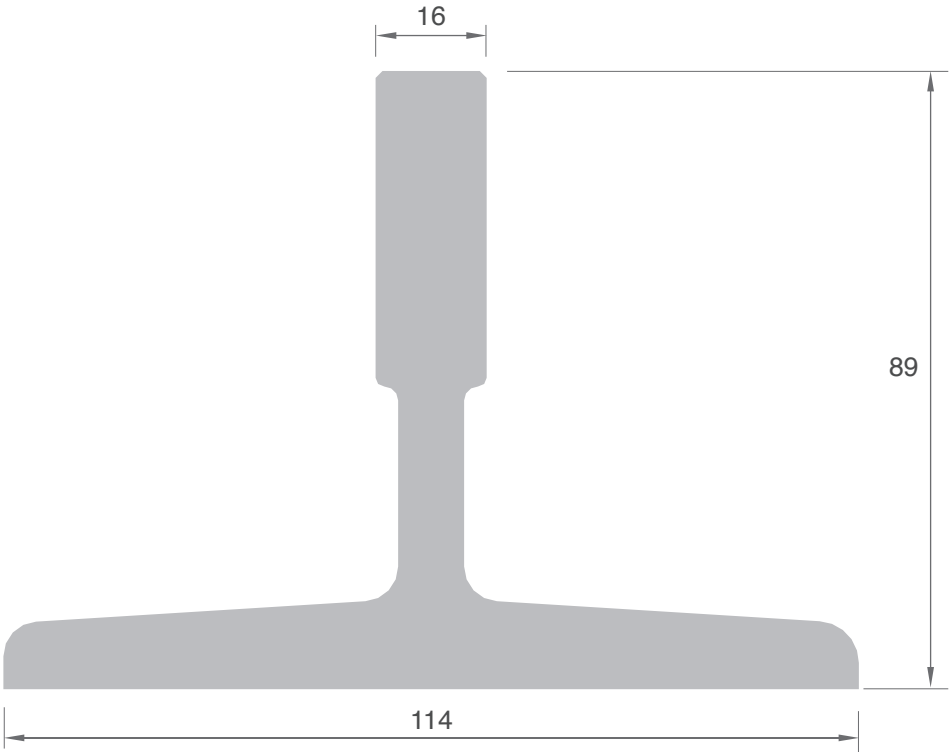


● SPECIFICATION

Unit Weight		12.04 (kg/m)
Standard Length		5,000 (mm)
Roughness	Long	$Ra \leq 3.2 \mu m$
	Trans.	$Ra \leq 6.3 \mu m$

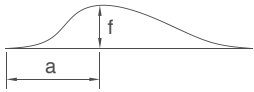
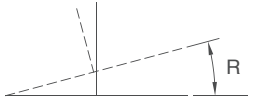
Straightness		f/5m	2.0mm
		f/0.5m	0.2mm
Twist		R	30'/m

Real Size

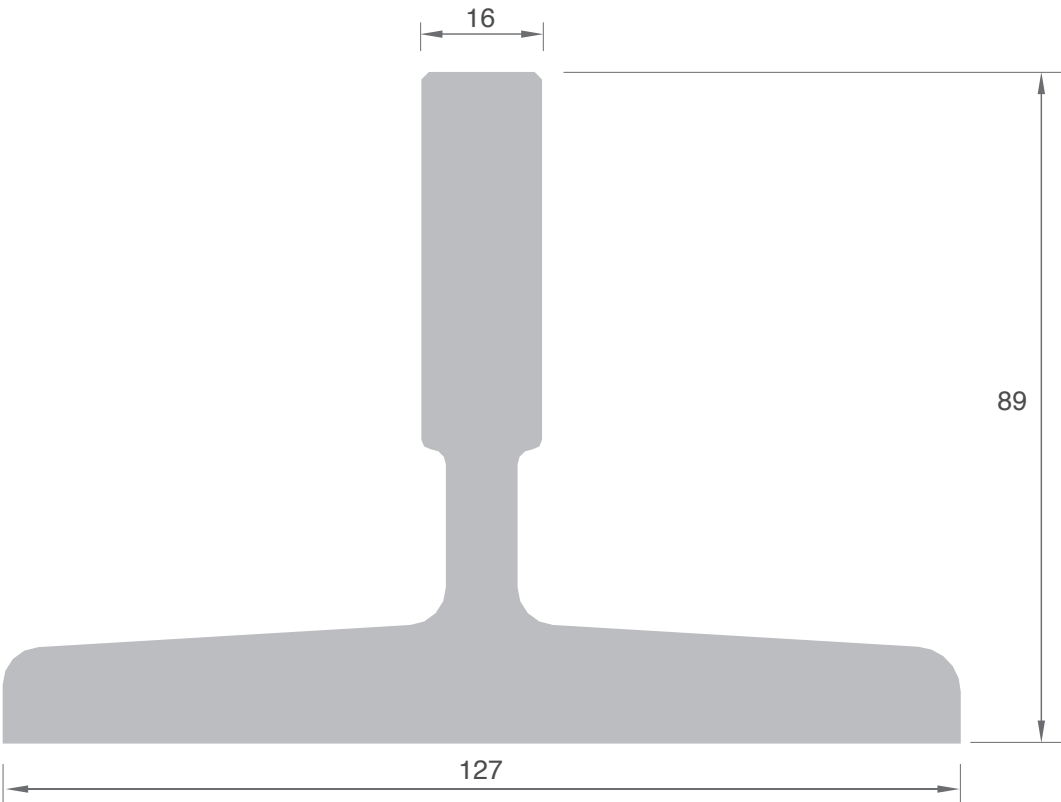


 SPECIFICATION

Unit Weight		16.78 (kg/m)
Standard Length		5,000 (mm)
Roughness	Long	$Ra \leq 3.2 \mu m$
	Trans.	$Ra \leq 6.3 \mu m$

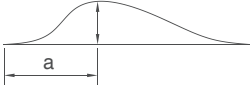
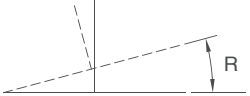
Straightness		f/5m	1.5mm
		f/0.5m	0.2mm
Twist		R	30'/m

Real Size

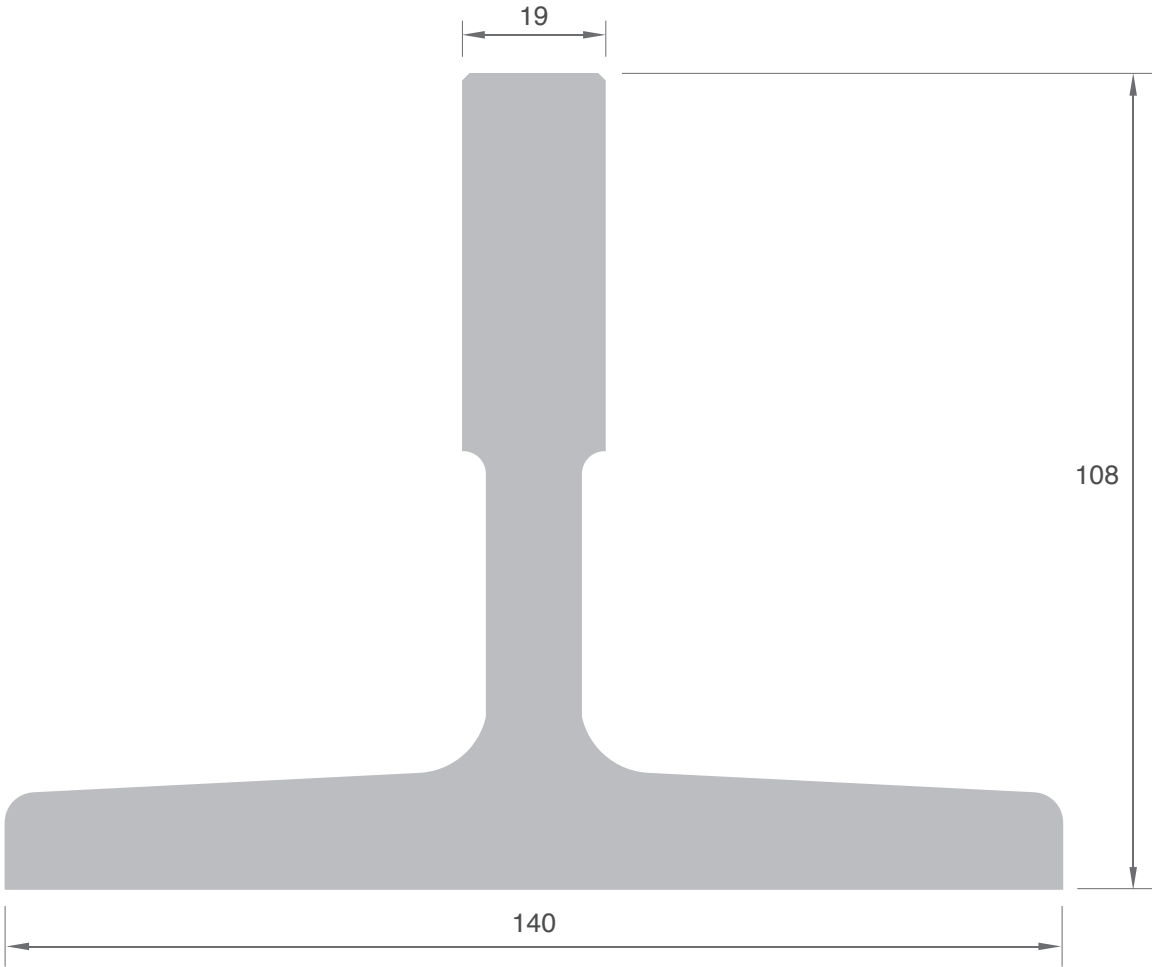


 SPECIFICATION

Unit Weight		22.18 (kg/m)
Standard Length		5,000 (mm)
Roughness	Long	$Ra \leq 3.2 \mu m$
	Trans.	$Ra \leq 6.3 \mu m$

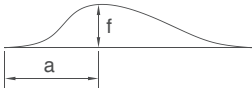
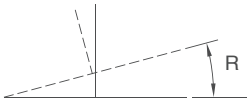
Straightness		f/5m	1.0mm
		f/0.5m	0.2mm
Twist		R	30'/m

Real Size

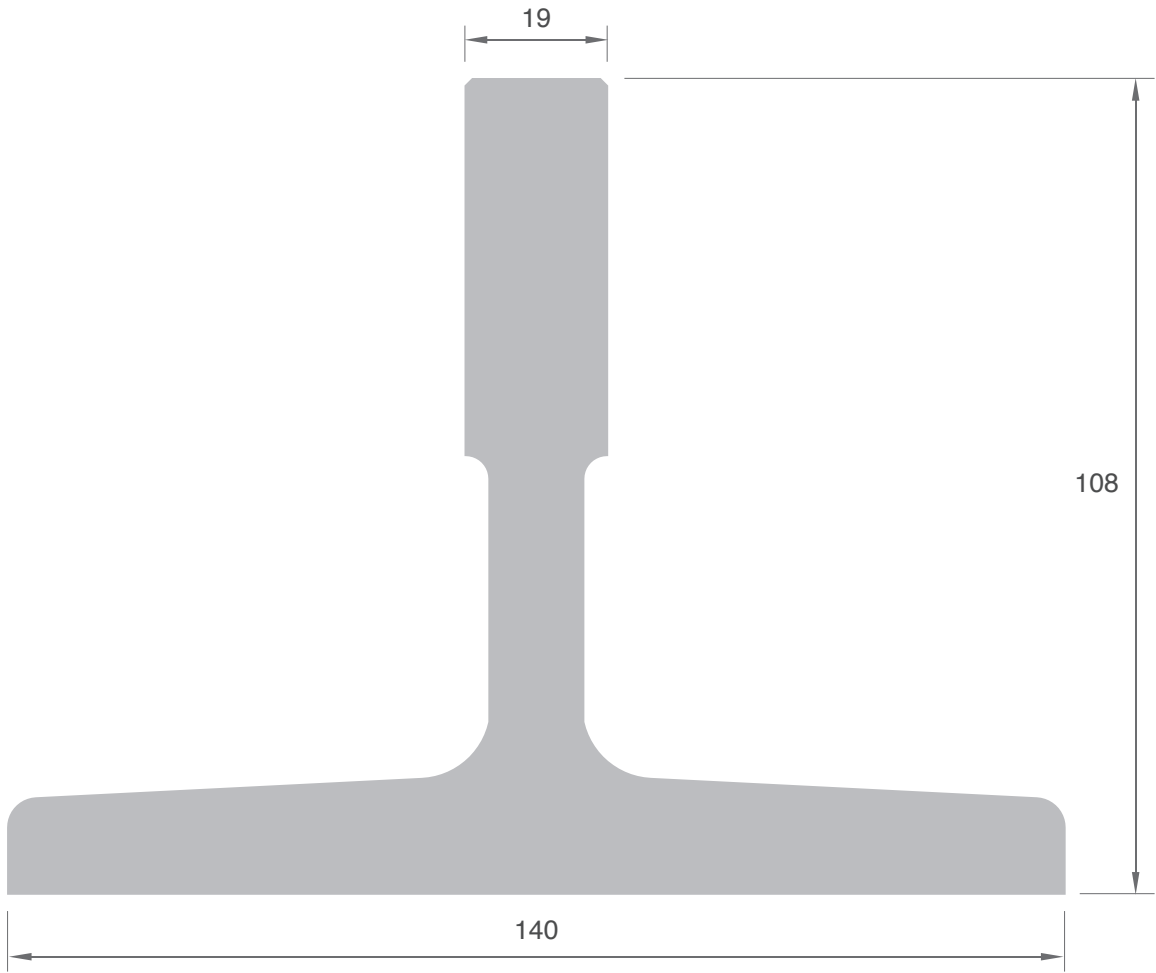


 SPECIFICATION

Unit Weight		27.42 (kg/m)
Standard Length		5,000 (mm)
Roughness	Long	$Ra \leq 3.2 \mu m$
	Trans.	$Ra \leq 6.3 \mu m$

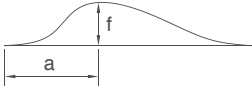
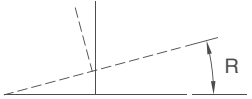
Straightness		f/5m	1.0mm
		f/0.5m	0.2mm
Twist		R	30'/m

Real Size

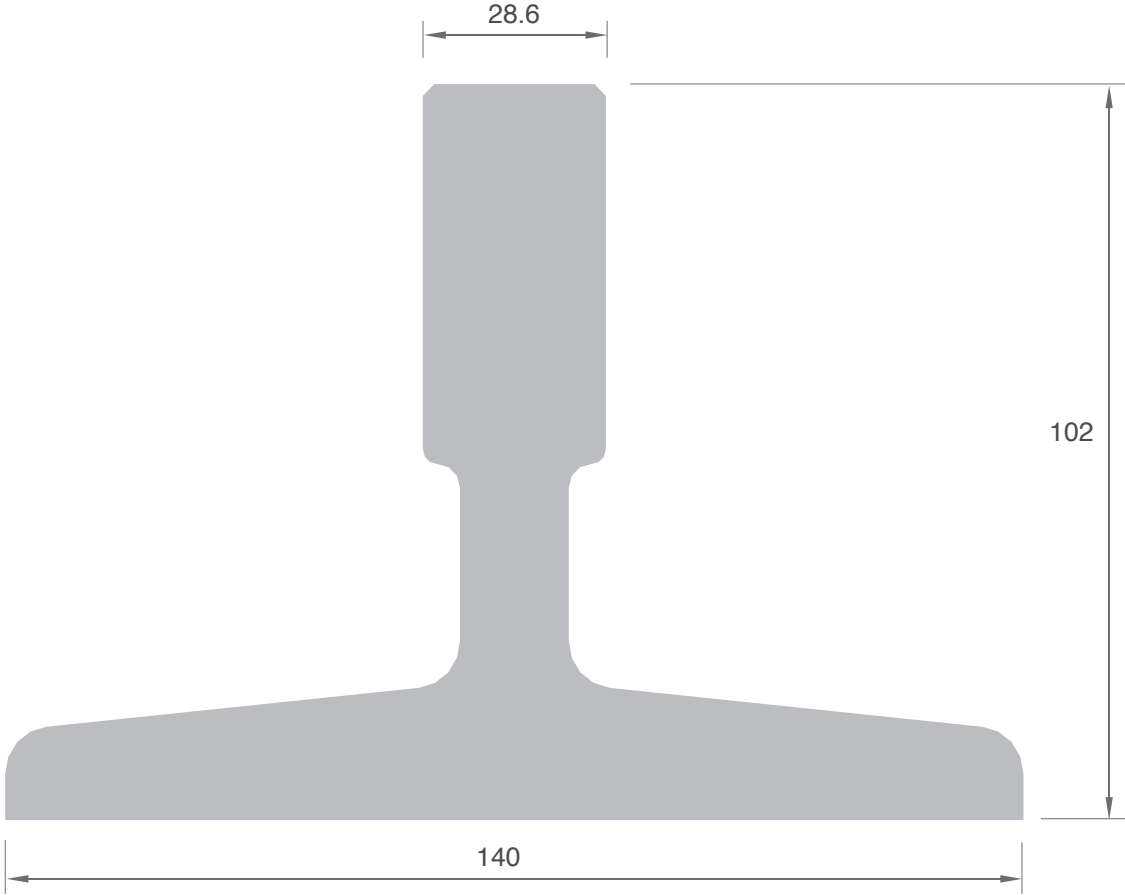


 SPECIFICATION

Unit Weight		27.42 (kg/m)
Standard Length		5,000 (mm)
Roughness	Long	$Ra \leq 1.6 \mu m$
	Trans.	$Ra \leq 3.2 \mu m$

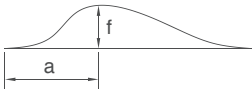
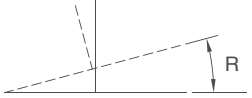
Straightness		f/5m	0.5mm
		f/0.5m	0.15mm
Twist		R	10'/m

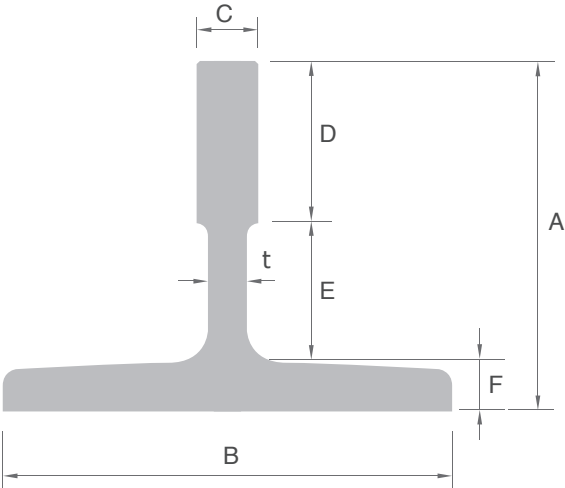
Real Size



 SPECIFICATION

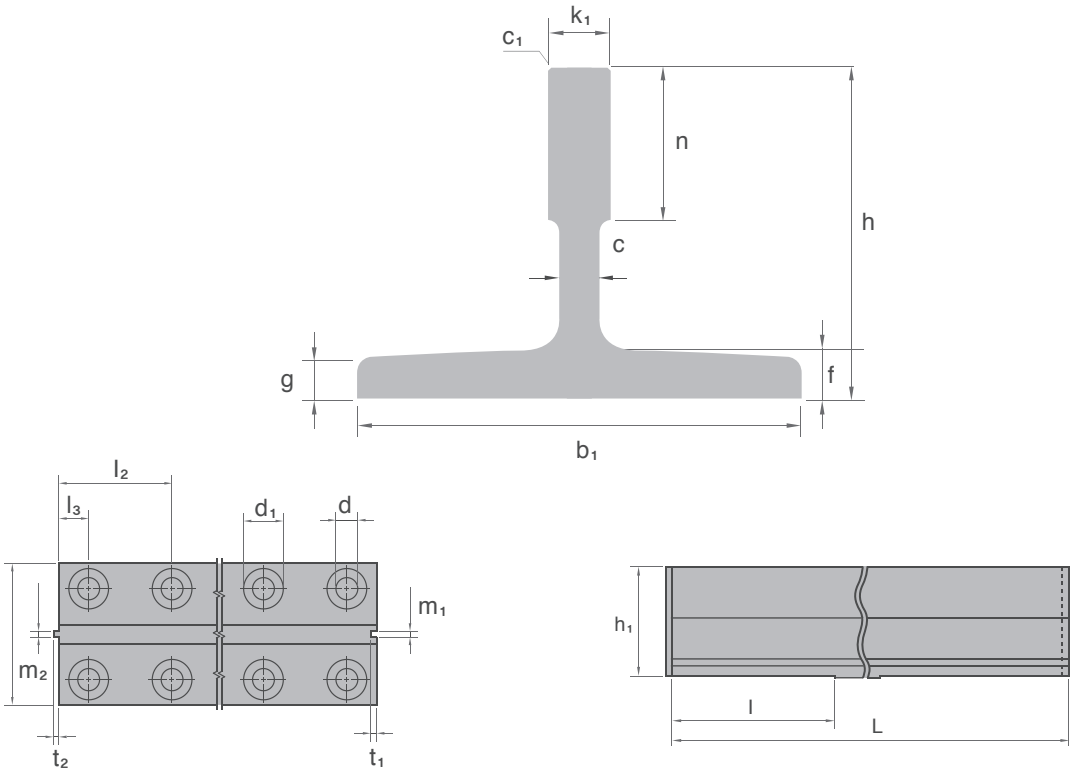
Unit Weight		34.1 (kg/m)
Standard Length		5,000 (mm)
Roughness	Long	$Ra \leq 1.6 \mu m$
	Trans.	$Ra \leq 3.2 \mu m$

Straightness		f/5m	0.5mm
		f/0.5m	0.15mm
Twist		R	10'/m



 DIMENSION

Class	Dimension							Area (cm ²)	Weight (kg/m)
	A	B	C	D	E	F	t		
JEAS 8K	58.0	78.0	13.0	28.0	21.5	8.5	7.0	10.9	8.55
JEAS 13K	63.5	89.0	19.0	33.0	19.0	11.5	9.5	16.7	13.1
JEAS 18K	90.5	114.0	19.0	39.0	40.0	11.0	9.5	22.3	17.5
JEAS 24K	90.5	127.0	19.0	51.5	23.0	16.0	9.5	30.2	23.7
JEAS 30K	110.0	140.0	22.5	52.5	41.5	16.0	10.5	37.8	29.7
ISO T89	63.5	89.0	19.0	33.0	19.0	11.5	9.5	16.69	13.1
ISO T127	90.5	127.0	19.0	51.5	23.0	16.0	9.5	30.18	23.7
ISO T140-1	110.0	140.0	22.5	52.5	41.5	16.0	10.5	37.8	29.7
37K	104.0	140.0	31.6	53.0	33.5	17.5	17.5	45.6	35.79

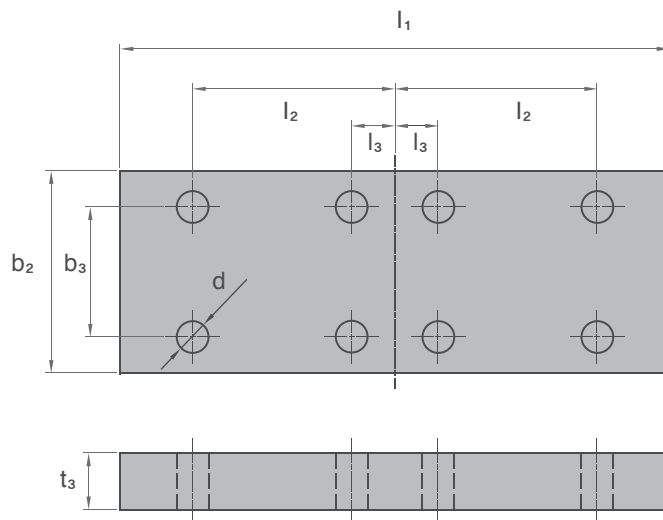


 DIMENSION

UNIT : mm

	b_1	h	h_1	l	k_1	n	c	f	g	m_1	m_2	t_1	t_2	d	d_1	L
Class	± 1.5	± 0.75	± 0.04	$\begin{smallmatrix} +10 \\ 0 \end{smallmatrix}$	$\begin{smallmatrix} 0 \\ -0.05 \end{smallmatrix}$	$\begin{smallmatrix} +3 \\ 0 \end{smallmatrix}$		± 0.75	± 0.75	$\begin{smallmatrix} +0.05 \\ +0.02 \end{smallmatrix}$	$\begin{smallmatrix} -0.02 \\ -0.05 \end{smallmatrix}$	± 0.2	± 0.2			± 2
8K	78	56	55	160	10	26	7.0	8.5	6.0	3	3	3.5	3	14.0	28	5,000
13K	89	62	61	160	16	32	9.5	11.5	7.5	4	4	4.5	4	14.0	28	
18K	114	89	88	160	16	38	9.5	11.0	8.0	6	6	5.5	5	17.5	34	
24K	127	89	88	160	16	50	9.5	16.0	12.5	6	6	5.5	5	17.5	34	
30K	140	108	107	193	19	50	12.7	15.9	12.7	5.98	6	5.5	5	21.0	40	5,000

Class	b_1	h	h_1	l	k_1	n	c	f	g	m_1	m_2	t_1	t_2	d	d_1	L
	± 1.5	± 0.75	± 0.03	$\begin{smallmatrix} +3 \\ 0 \end{smallmatrix}$	$\begin{smallmatrix} +0.05 \\ 0 \end{smallmatrix}$	$\begin{smallmatrix} +3 \\ 0 \end{smallmatrix}$		± 0.75	± 0.75	$\begin{smallmatrix} +0.03 \\ 0 \end{smallmatrix}$	$\begin{smallmatrix} 0 \\ -0.03 \end{smallmatrix}$	± 0.1	± 0.1			$\begin{smallmatrix} +2 \\ 0 \end{smallmatrix}$
High speed 30K	140	108	107	193	19	50	12.7	15.9	12.7	6	5.98	5.5	5	21	40	5,000
High speed 37K	140	102	101	193	28.6	51	17.5	17.5	14.5	6.4	6.37	7.14	6.35	21	40	5,000



DIMENSION FOR FISH PLATE

UNIT : mm

Class 규격	l_1	b_2	l_2	l_3	b_3	t_3	d	Bolt, Nut	Spring Washer
	± 3	± 3	± 0.2	± 0.2	± 0.2	± 0.5	± 0.5		
8K	270	78	114	38	50	10	14.0	M12×40	A12
13K	280	89	114	38	56	15	14.0	M12×50	A12
18K	280	114	114	38	71	20	17.5	M16×65	A16
24K	280	127	114	38	80	20	17.5	M16×65	A16
30K	380	140	140	30	90	39	22.0	M20×65	A20
High speed 30K	380	140	140	30	90	39	22.0	M20×65	A20
High speed 37K	380	140	140	30	90	39	22.0	M20×80	A20

CHEMICAL COMPOSITION & MECHANICAL PROPERTIES

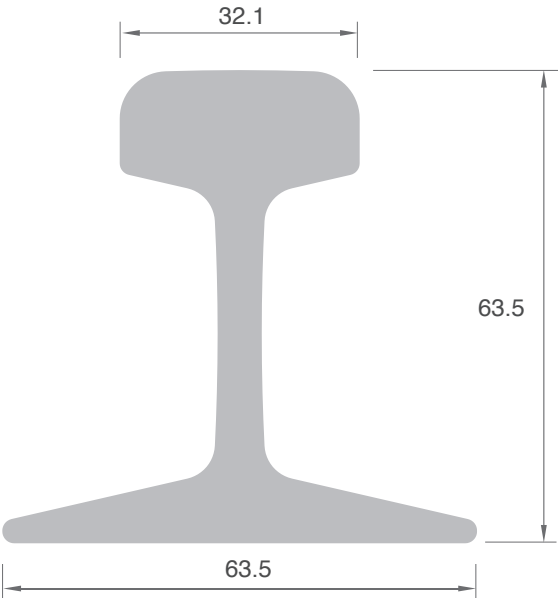
Class	Chemical Composition					Mechanical Properties		
	C (%)	Si (%)	Mn (%)	P (%)	S (%)	Yield Point (N/mm²)	Tensile Strength (N/mm²)	Elongation (%)
8K	0.08 ~ 0.30	Max 0.35	0.30 ~ 0.75	Max 0.050	Max 0.050	255 Min	402 Min	23.0 Min
13K								
18K								
24K								
30K								
High speed 30K								
High speed 37K								

SECTIONAL PROPERTIES

Class	Moment to Inertia (cm⁴)		Radius of Gyration (cm)		Modulus of Section (cm³)	
	Ix	Iy	ix	iy	Zx	Zy
8K	29.90	26.10	1.74	1.63	7.56	6.71
13K	59.50	50.40	1.96	1.81	14.30	11.30
18K	179.00	109.00	2.93	2.29	29.70	19.10
24K	199.00	277.00	2.65	2.83	35.70	31.10
30K	403.00	311.00	3.40	2.99	53.10	44.50
High speed 30K	403.00	311.00	3.40	2.99	53.10	44.50
High speed 37K	457.00	358.00	3.24	2.87	68.00	51.20

LIGHT
RAIL
/RAIL

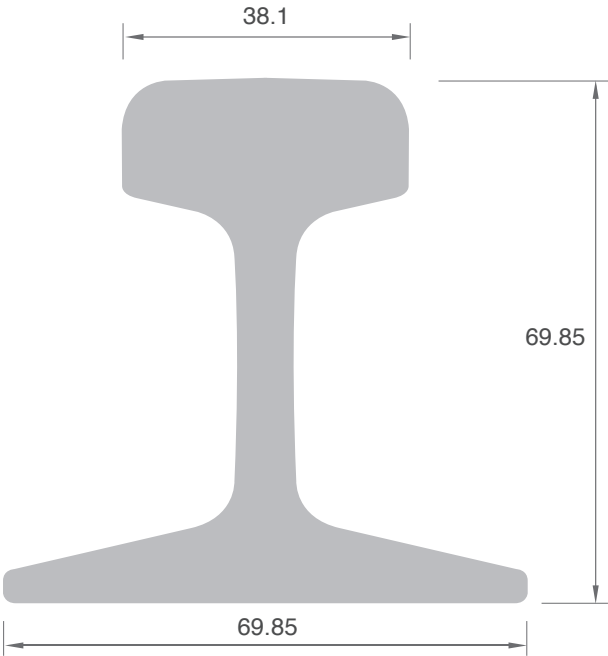
Real Size



 SPECIFICATION

Unit Weight	8.94 (kg/m)	Standard Drilling	15.9 (mm)
Standard Length	6,000 (mm) / 5,500 (mm) / 6,096 (20FT.)	Tensile Strength	Min 569 (N/mm ²)
Sectional Area	11.39 (cm ²)	Elongation	Min 12.0 (%)

Real Size

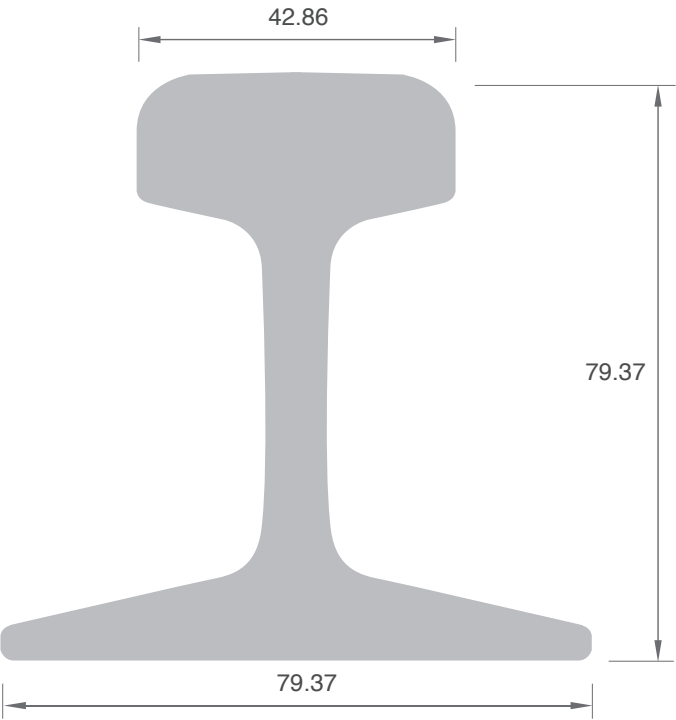


 SPECIFICATION

Unit Weight	12.2 (kg/m)
Standard Length	10,000 (mm) / 5,000 (mm) / 6,096 (20FT.) / 9,144 (30FT.) / 10,058 (33FT.) / 12,192 (40FT.)
Sectional Area	15.54 (cm ²)

Standard Drilling	15.9 (mm)
Tensile Strength	Min 569 (N/mm ²)
Elongation	Min 12.0 (%)

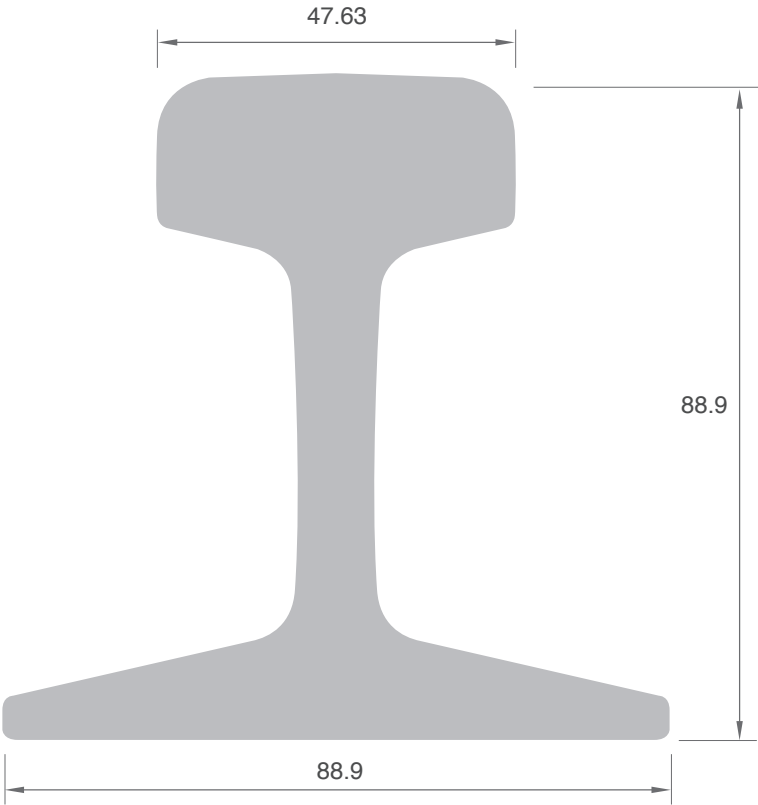
Real Size



 SPECIFICATION

Unit Weight	15.2 (kg/m)	Standard Drilling	19.05 (mm)
Standard Length	10,000 (mm) / 5,000 (mm) / 6,096 (20FT.) / 9,144 (30FT.) / 10,058 (33FT.) / 12,192 (40FT.)	Tensile Strength	Min 569 (N/mm ²)
Sectional Area	19.33 (cm ²)	Elongation	Min 12.0 (%)

Real Size

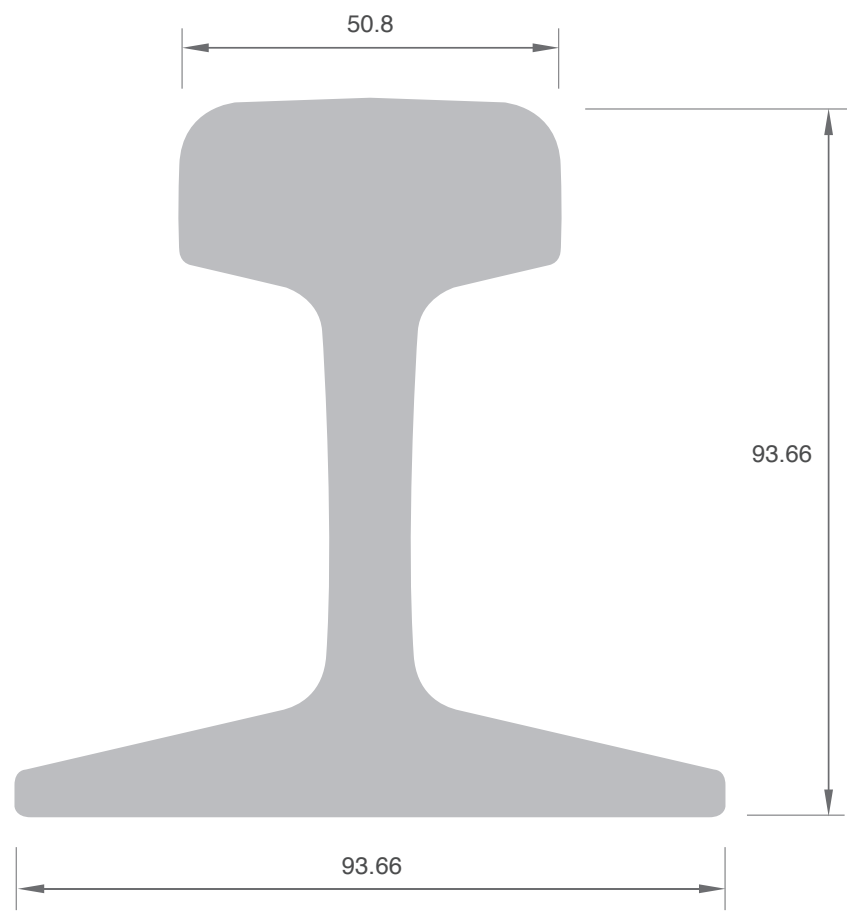


 SPECIFICATION

Unit Weight	19.80 (kg/m)
Standard Length	10,000 (mm) / 5,000 (mm) / 6,096 (20FT.) / 9,144 (30FT.) / 10,058 (33FT.) / 12,192 (40FT.)
Sectional Area	25.40 (cm²)

Standard Drilling	22.22 (mm)
Tensile Strength	Min 569 (N/mm²)
Elongation	Min 12.0 (%)

Real Size

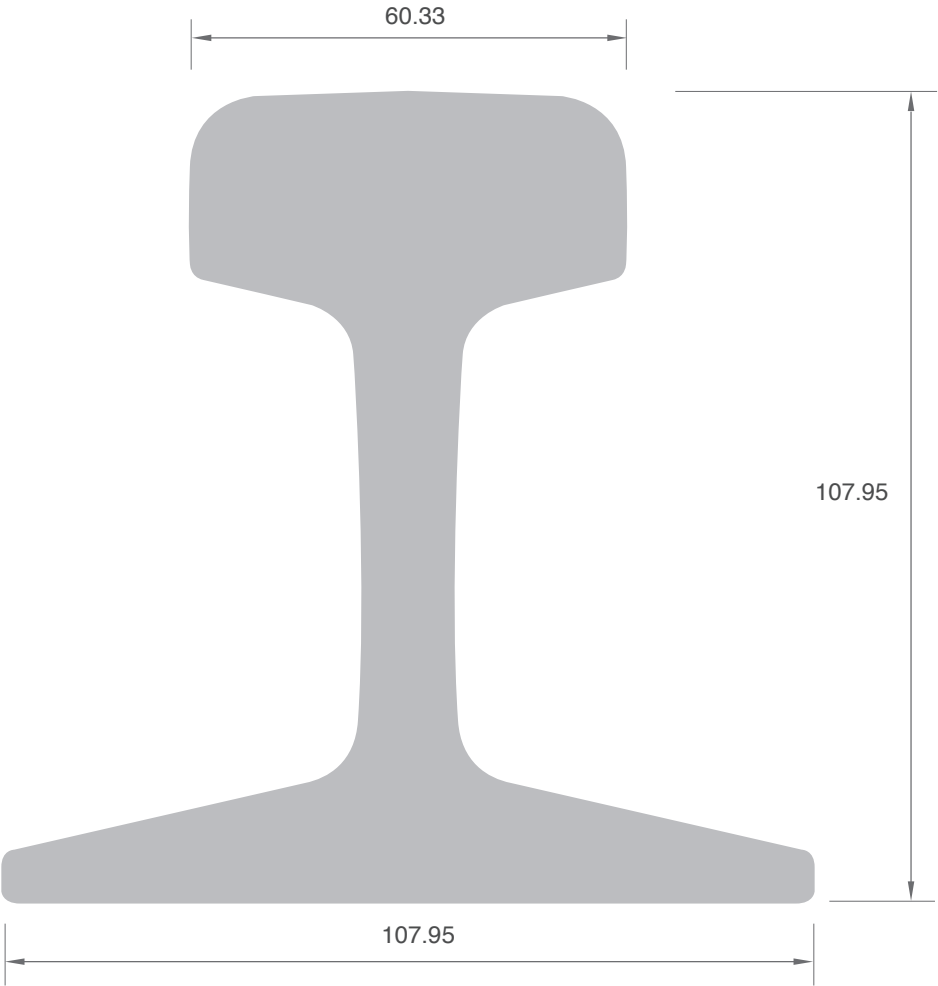


 SPECIFICATION

Unit Weight	22.3 (kg/m)
Standard Length	10,000 (mm) / 5,000 (mm) / 6,096 (20FT.) / 9,144 (30FT.) / 10,058 (33FT.) / 12,192 (40FT.)
Sectional Area	28.39 (cm ²)

Standard Drilling	22.22 (mm)
Tensile Strength	Min 637 (N/mm ²)
Elongation	Min 10.0 (%)

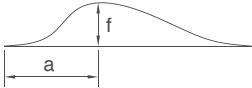
Real Size



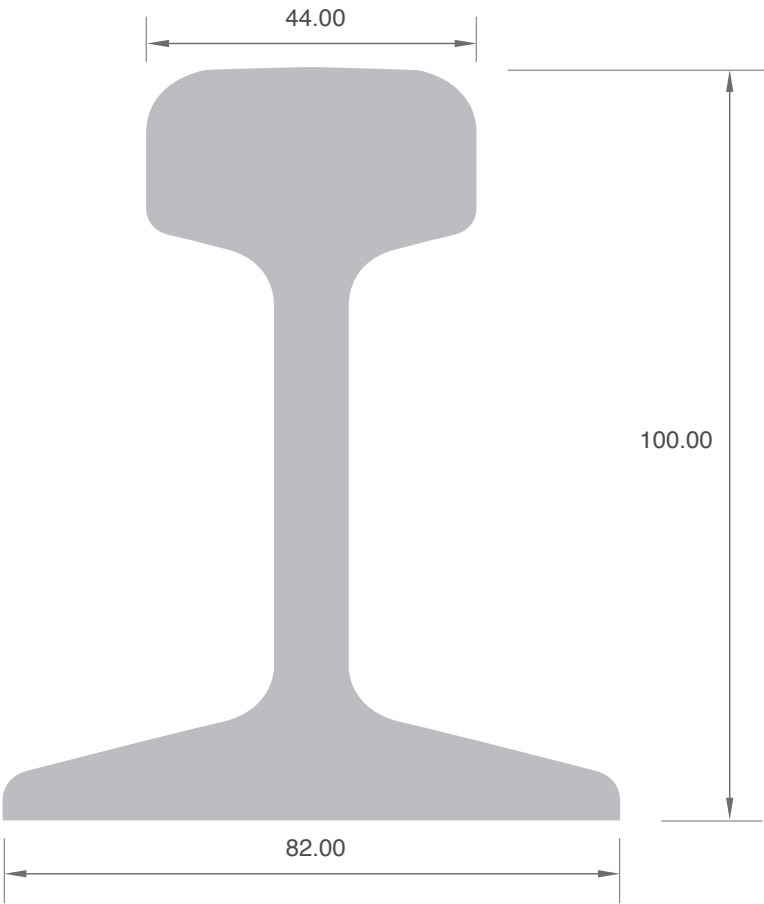
 SPECIFICATION

Unit Weight	30.1 (kg/m)
Standard Length	10,000 (mm) / 5,000 (mm) / 6,096 (20FT.) / 9,144 (30FT.) / 10,058 (33FT.) / 12,192 (40FT.)
Sectional Area	38.26 (cm ²)

Standard Drilling	25.40 (mm)
Tensile Strength	Min 690 (N/mm ²)
Elongation	Min 9.0 (%)
Straightness	15mm / 10m



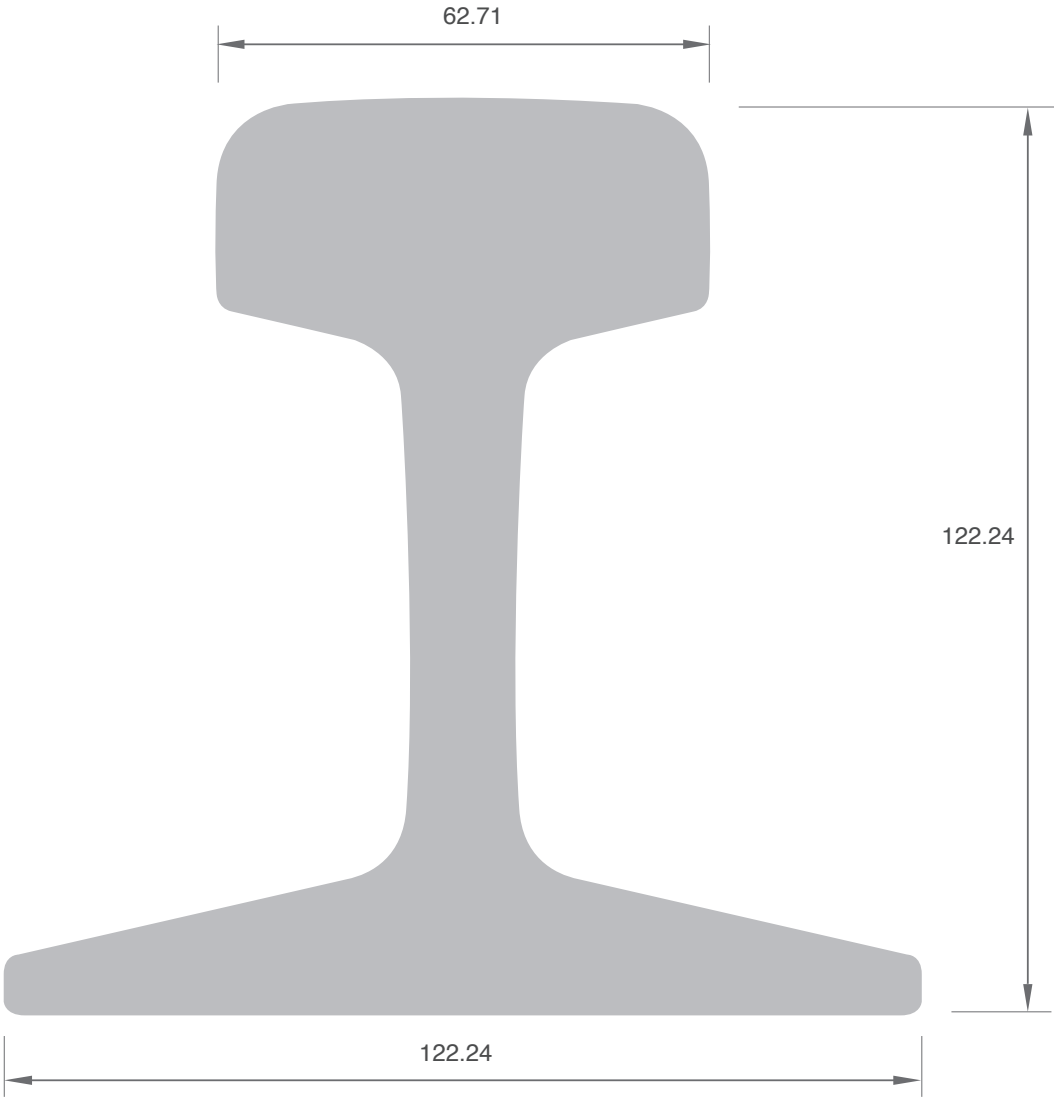
Real Size



 SPECIFICATION

Unit Weight	19.8 (kg/m)	Standard Drilling	44.6 (mm)
Standard Length	10,000 (mm)	Tensile Strength	Min 580 (N/mm²)
Sectional Area	25.28 (cm²)	Elongation	Min 12.0 (%)

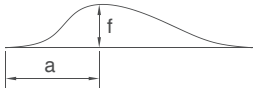
Real Size

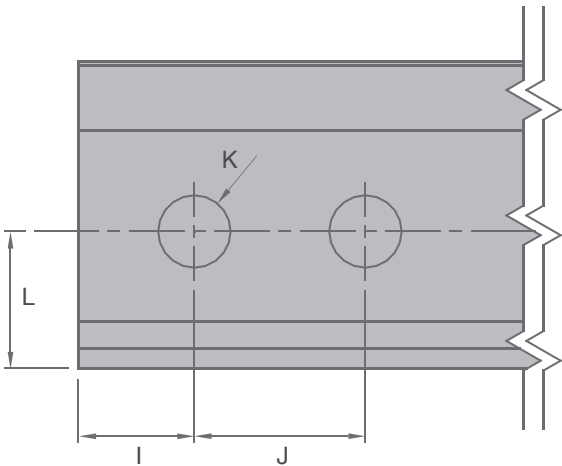
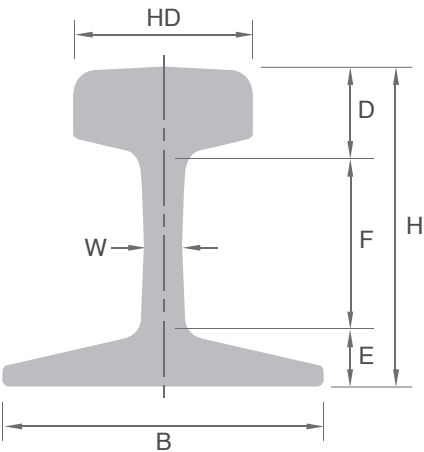


 SPECIFICATION

Unit Weight	37.2 (kg/m)
Standard Length	10,000 (mm)
Sectional Area	47.3 (cm ²)

Standard Drilling	25.40 (mm)
Tensile Strength	Min 690 (N/mm ²)
Elongation	Min 9.0 (%)
Straightness	10mm / 10m





DIMENSION

UNIT : mm

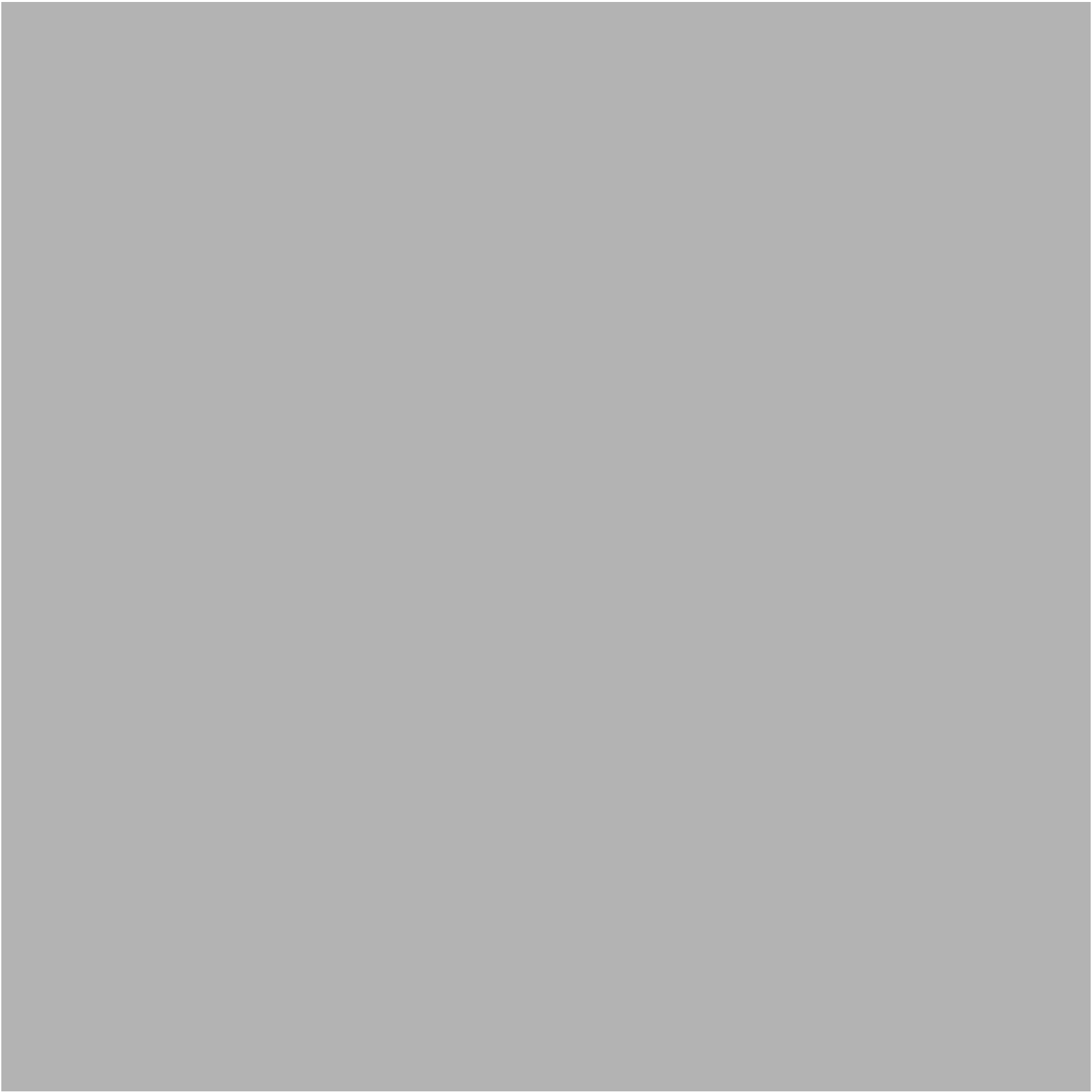
Class	Standard Sectional Dimension										
	H	B	HD	D	E	F	L	J	I	K	W
9kg (18lbs)	63.50	63.50	32.10	17.48	35.72	10.30	28.16	101.6	50.8	15.90	5.90
12kg (25lbs)	69.85	69.85	38.10	19.85	37.70	12.30	31.15	101.6	50.8	15.90	7.54
15kg (30lbs)	79.37	79.37	42.86	22.22	43.65	13.50	35.32	101.6	50.8	19.05	8.33
20kg (40lbs)	88.90	88.90	47.62	25.79	47.23	15.88	39.69	127.0	63.5	22.22	9.92
22kg (45lbs)	93.66	93.66	50.80	26.99	50.00	16.67	41.67	127.0	63.5	22.22	10.70
30kg (60lbs)	107.95	107.95	60.33	30.95	57.55	19.45	42.22	127.0	63.5	23.80	12.30
37kg (75lbs)	122.24	122.24	62.71	36.12	21.43	64.69	53.78	127.0	60.5	25.40	13.49
DIN S-20	100.00	82.00	44.00	27.00	16.20	56.80	44.60	90.0	35.0	22.00	10.00

● CHEMICAL COMPOSITION & MECHANICAL PROPERTIES

Class	Chemical Composition				
	C (%)	Si (%)	Mn (%)	P (%)	S (%)
9kg (18lbs)	0.40 ~ 0.60	Max 0.40	0.50 ~ 0.90	Max 0.045	Max 0.050
12kg (25lbs)					
15kg (30lbs)					
20kg (40lbs)					
22kg (45lbs)	0.45 ~ 0.65	Max 0.40	0.50 ~ 0.90	Max 0.045	Max 0.050
30kg (60lbs)	0.50 ~ 0.70	0.10 ~ 0.35	0.60 ~ 0.95	Max 0.045	Max 0.050
37kg (75lbs)	0.55 ~ 0.70				
DIN S-20	0.40 ~ 0.60	Max 0.40	0.50 ~ 0.90	Max 0.045	Max 0.050

● SECTIONAL PROPERTIES

Class	Weight (kg/m)	Sectional Area(cm²)	Moment of Inertia (cm⁴)	Radius of Gyration (cm)	Modulus of Section (cm³)
9kg (18lbs)	8.94	11.39	62.41	2.33	19.10
12kg (25lbs)	12.20	15.54	98.82	2.51	27.60
15kg (30lbs)	15.20	19.33	156.10	2.83	38.60
20kg (40lbs)	19.80	25.40	272.20	3.28	58.80
22kg (45lbs)	22.30	28.39	339.00	3.45	69.60
30kg (60lbs)	30.10	38.26	606.00	3.98	116.00
37kg (75lbs)	37.20	47.30	952.00	-	164.00
DIN S-20	19.80	25.28	346.00	-	66.80

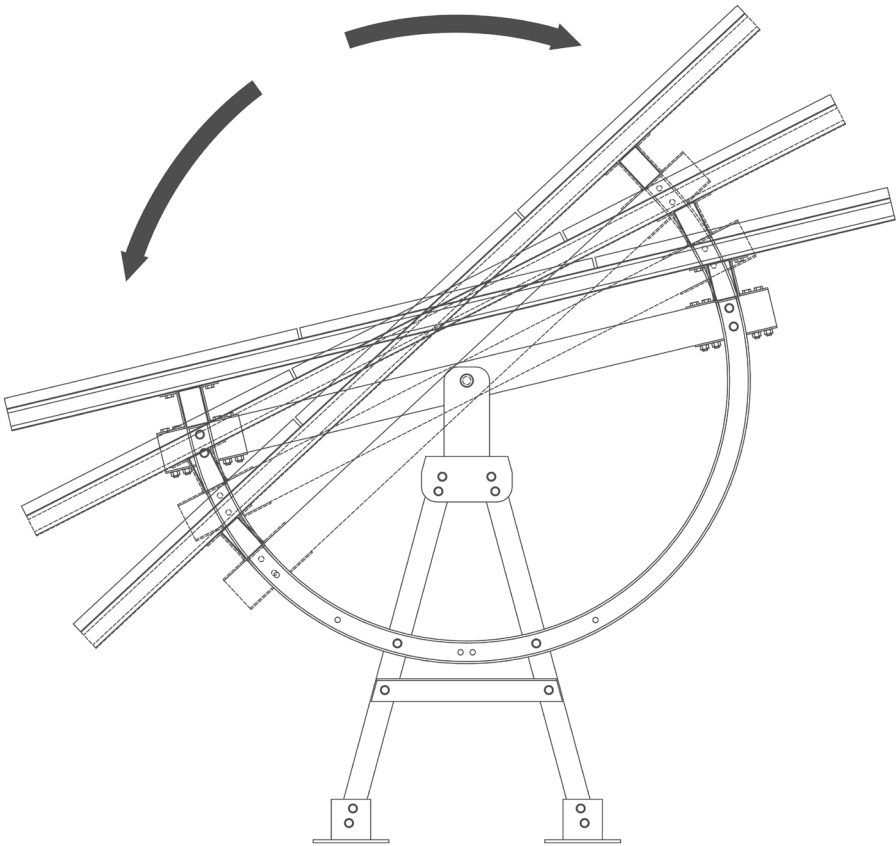




TRACKER

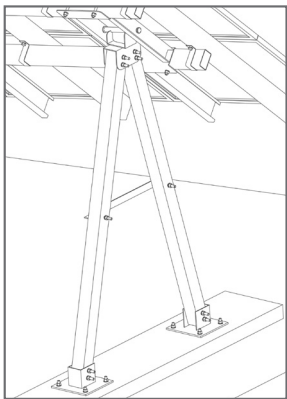
5-1

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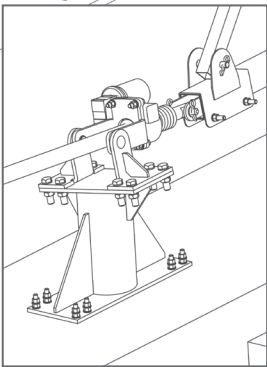
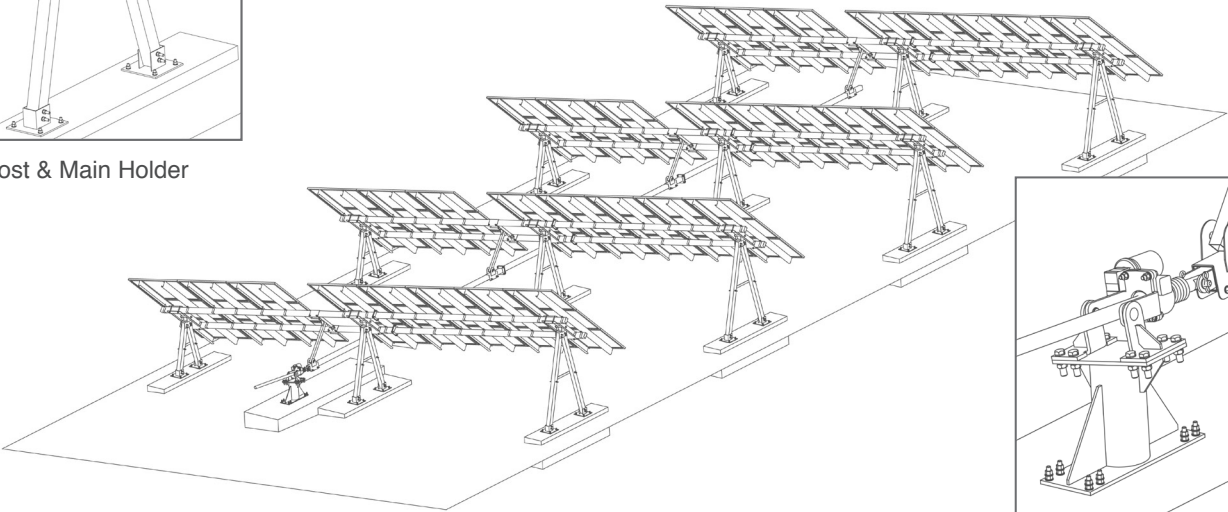


 SPECIFICATION

Type	Fixed type(Tilt fixed)	Seasonally adjustable type according to solar altitude
Model	IK-FT1	Improve power efficiency by manual adjustable type
Size	Variable according to installed conditions	Design and maufacturing considering plant site, solar incidence angle
		Structure analysis and material selection accoring to load(wind load, snow load, etc.)



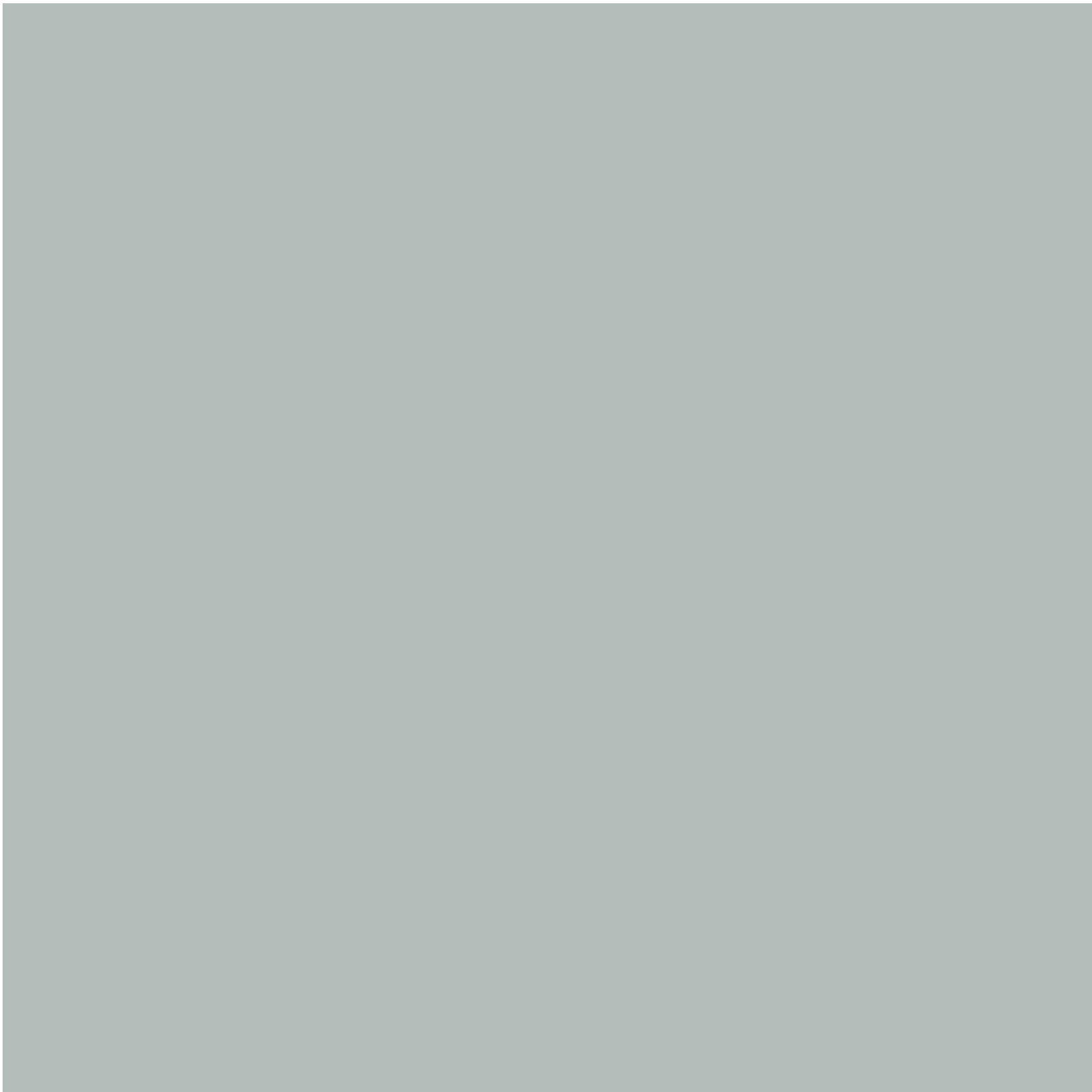
■ Post & Main Holder



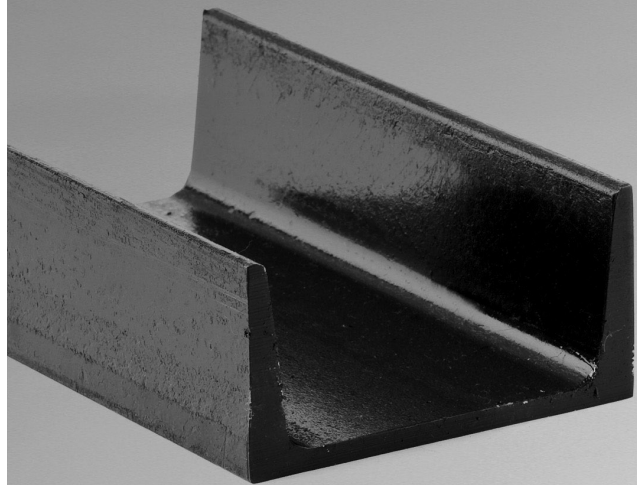
■ Tracking System

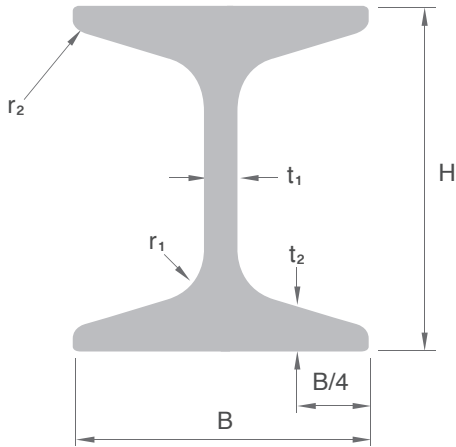
○ SPECIFICATION

Type	Single axis type(Single axis tracking)	Improve power generation efficiency by tracking system
Model	IK-SA1	Precise tracking location of the sun based on astronomical meteorological data
Size	Variable according to installed conditions	Design and maufacturing considering plant site, solar incidence angle
		Structure analysis and material selection according to load(wind load, snow load, etc.)



SPECIAL SHAPE



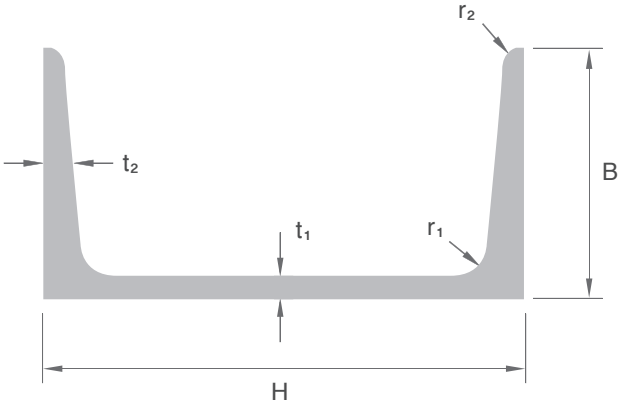


DIMENSION AND SECTIONAL PROPERTIES

Standard Sectional Dimension (mm)					Sectional Area (cm ²)	Unit Weight (kg/m)	Moment of Inertia (cm ⁴)		Radius of Gyration (cm)		Modulus of Section (cm ³)	
H×B	t ₁	t ₂	r ₁	r ₂	A	W	I _x	I _y	i _x	i _y	Z _x	Z _y
100×80	9	12.5	13	4	26.4	20.7	403	80.5	3.91	1.75	80.7	20.1

CHEMICAL COMPOSITION & MECHANICAL PROPERTIES

Class	Chemical Composition					Mechanical Properties		
	C (%)	Si (%)	Mn (%)	P (%)	S (%)	Yield Point (N/mm ²)	Tensile Strength (N/mm ²)	Elongation (%)
				Max 0.05	Max 0.05		Min 490	Min 20

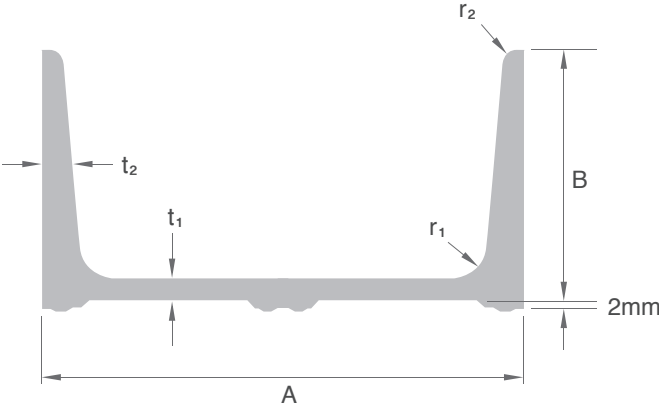


DIMENSION AND SECTIONAL PROPERTIES

Standard Sectional Dimension (mm)					Sectional Area (cm ²)	Unit Weight (kg/m)	Moment of Inertia (cm ⁴)		Radius of Gyration (cm)		Modulus of Section (cm ³)	
H×B	t ₁	t ₂	r ₁	r ₂	A	W	I _x	I _y	i _x	i _y	Z _x	Z _y
75×40	5	7	8	4	8.818	6.92	75.3	12.2	2.92	1.17	20.1	4.47
100×50	5	7.5	8	4	11.92	9.36	188	26	3.97	1.48	37.6	7.52
125×65	6	8	8	4	17.11	13.4	424	61.8	4.98	1.9	67.8	13.4

CHEMICAL COMPOSITION & MECHANICAL PROPERTIES

Class	Chemical Composition					Mechanical Properties		
	C (%)	Si (%)	Mn (%)	P (%)	S (%)	Yield Point (N/mm ²)	Tensile Strength (N/mm ²)	Elongation (%)
				Max 0.05	Max 0.05	Min 245	400 ~ 510	t1≤5 Min 21 5<t1≤16 Min 17



DIMENSION AND SECTIONAL PROPERTIES

Standard Sectional Dimension (mm)					Sectional Area(cm ²)	Unit Weight (kg/m)	Moment of Inertia (cm ⁴)	Modulus of Section (cm ³)
A×B	t ₁	t ₂	r ₁	r ₂				
125×65	6	8	8	4	17.31	14.5	68.40	14.61

CHEMICAL COMPOSITION & MECHANICAL PROPERTIES

Class	Chemical Composition					Mechanical Properties		
	C (%)	Si (%)	Mn (%)	P (%)	S (%)	Yield Point (N/mm ²)	Tensile Strength (N/mm ²)	Elongation (%)
				Max 0.05	Max 0.05	Min 245	400 ~ 510	Min 17

● GENERAL DECK PLATES

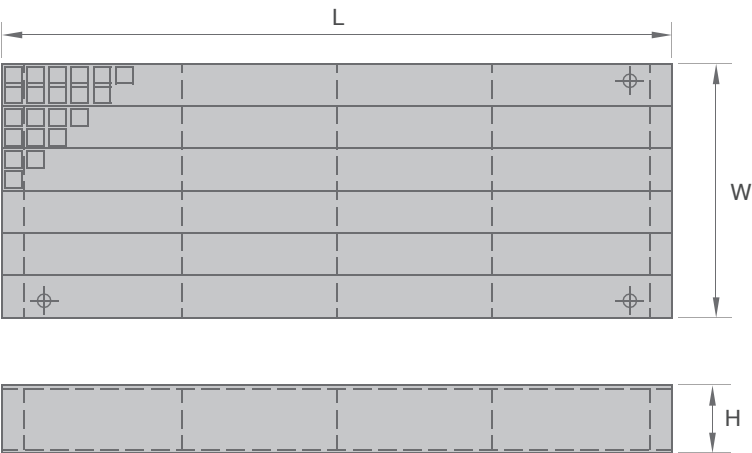
Feature	· Easy constructability due to standard specification · High quality & durability and easy transportation & perseverance
Use	· Usable for subway construction surface · Facility for Weakness of ground · Bridge construction · Civil Engineering construction
BPN	BPN : 30 ~ 40

● MMA COATING DECK PLATES

Feature	· MMA coating treatment gives excellence in resisting slipperiness, especially in heavy rain or snow, and does not absorb moisture · Low noise and vibration
Use	· Usable for subway construction surface · Facility for weakness of ground · Bridge construction · Civil engineering construction · Special areas requisite for slippery cautions
BPN	BPN : 60 ~ 64 / General Deck Plate & Asphalt (BPN: 20 ~ 38)

● GRATING DECK PLATES

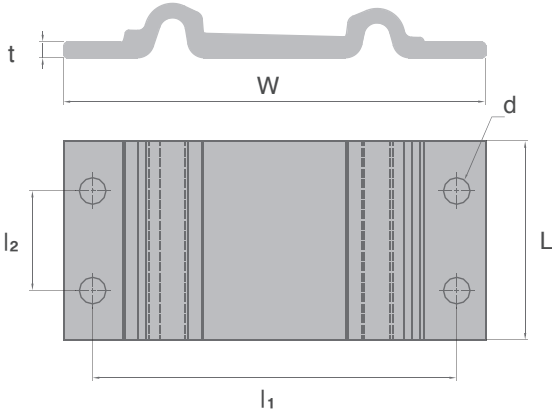
Feature	· On construction site, total or partial construction is capable for ventilation of polluted air · CX net can be installed over the product to prevent falling of hazard material through air hole
Use	· Usable for subway construction surface · Facility for weakness of ground · Bridge construction · Civil engineering construction · Special areas requisite for lighting and ventilation
BPN	BPN : 60 ~ 70



● DIMENSION AND SECTIONAL PROPERTIES

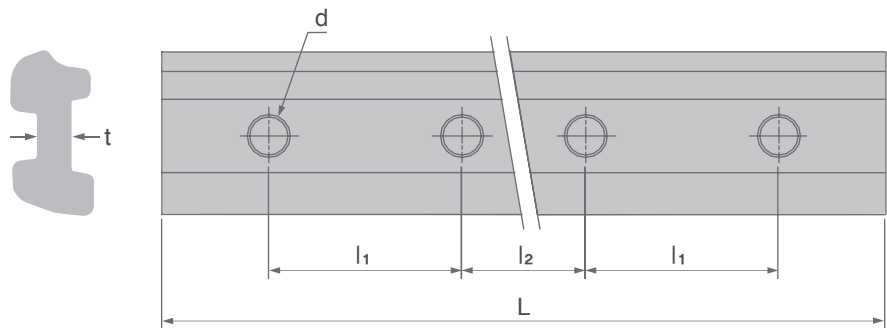
TYPE	Standard Sectional Dimension (mm)			Covering Area (m ²)	Unit Weight (kg/piece)	Sectional Area(cm ²)	Moment of Inertia (cm ⁴)	Modulus of Section (cm ³)
	W	L	H					
General (I - B - 750)	750	1,990	200	1.5	300	138.06	6.413	443
MMA Coating (TDT - 750)	750	1,990	204	1.5	300	138.06	6.413	443
Grating (DSG - 750)	750	1,990	200	1.5	285	138.06	6.413	443

※ Packing Material : Polymer concrete(MMA), Packing Thickness : T = 4mm



● BASE PLATE

Standard Sectional Dimension (mm)						Chemical Composition		Mechanical Properties		
W	L	l ₂	l ₁	d	t	P	S	Yield Point (N/mm ²)	Tensile Strength (N/mm ²)	Elongation (%)
381 +2 0	180 ±1.5	90 ±0.5	329 ±0.5	24 +1.5 0	14 ±0.6	Max 0.05	Max 0.05	Min 245	400 ~ 510	Min 17



FISH PLATE

Standard Sectional Dimension (mm)					Chemical Composition					Mechanical Properties		Bend Test
L	t	d	l ₁	l ₂	C	Si	Mn	P	S	Yield Point (N/mm ²)	Tensile Strength (N/mm ²)	
560 ±3	20 ±0.5	24 +1.0 -0.5	130 ± 1	160 ±1	0.35 ~ 0.53	0.15 ~ 0.35	0.6 ~ 0.9	Max 0.03	Max 0.035	Min 569	Min 15.0	90°

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